

NAME OF MY PROJECT

**M-Training Toolkit on
Polio / RI for CMCs
in
CoreGroup Polio Project.**

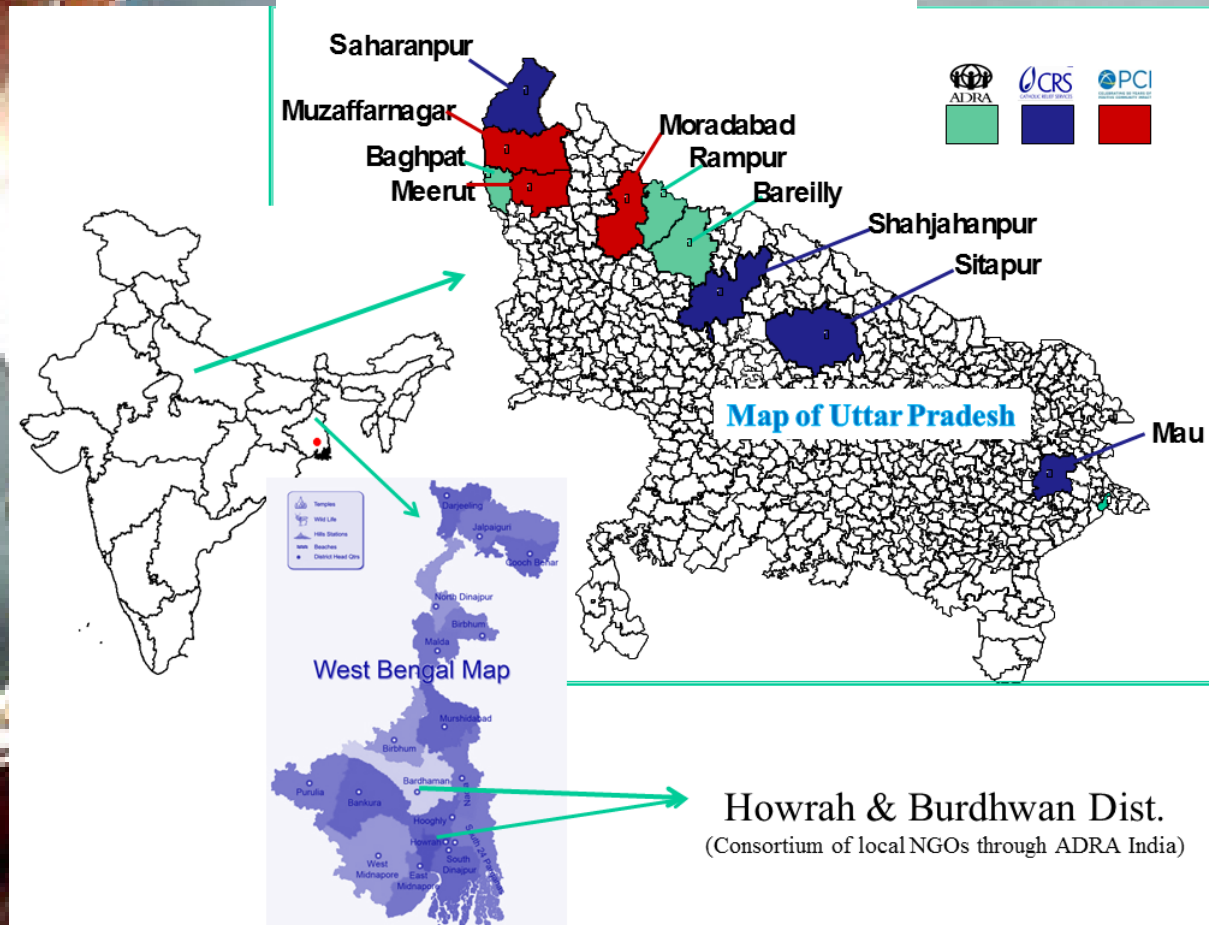
During my industrial training, I am working with ZMQ in the Mobile for Development(M4D) group.

In the M4D group, I am part of its CoreGroup-ZMQ Polio project.



**M-Health
Solution**

CoreGroup Polio Project Brief



Howrah & Burdwan Dist.
(Consortium of local NGOs through ADRA India)

- PCI** - Dist. of **Meerut**, **Moradabad** and **Muzzafarnagar** in U.P.
- ADRA** - Dist. of **Baghpat**, **Bareilly** and **Rampur** in U.P. (Howrah & Burdwan - W.B.)
- CRS** - Dist. of **Saharanpur**, **Shahjahanpur** and **Sitapur** in U.P.

CGPP: Implementation Structure



SA – Super Administration

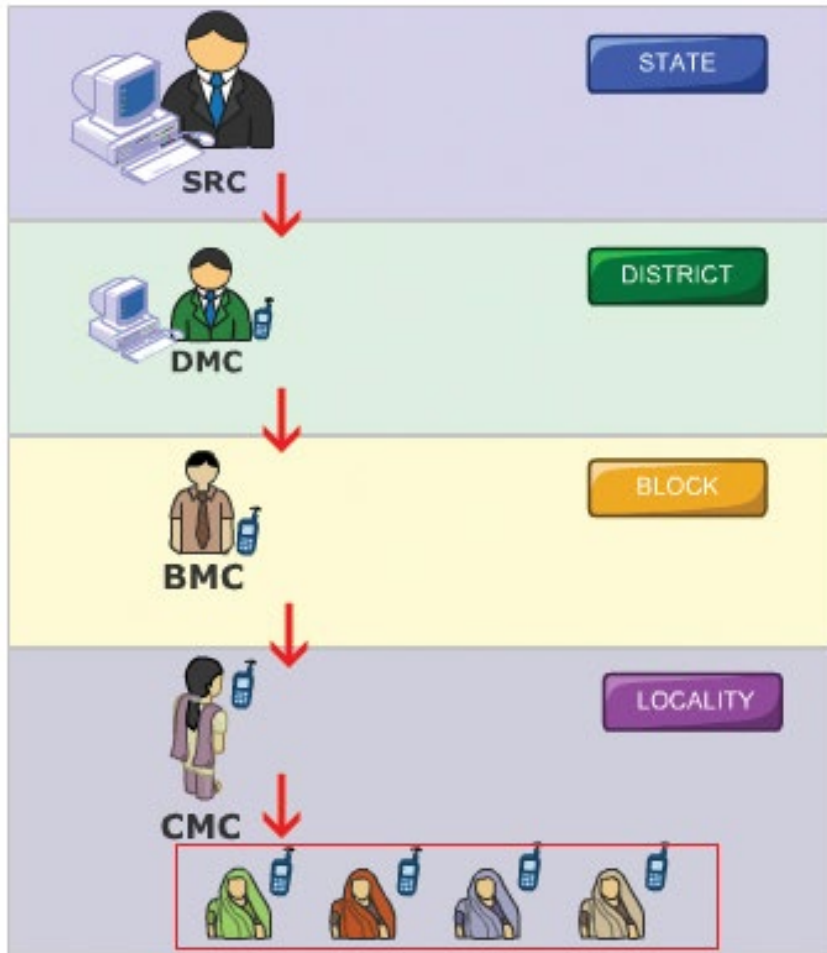
PL – Partners Management Layer

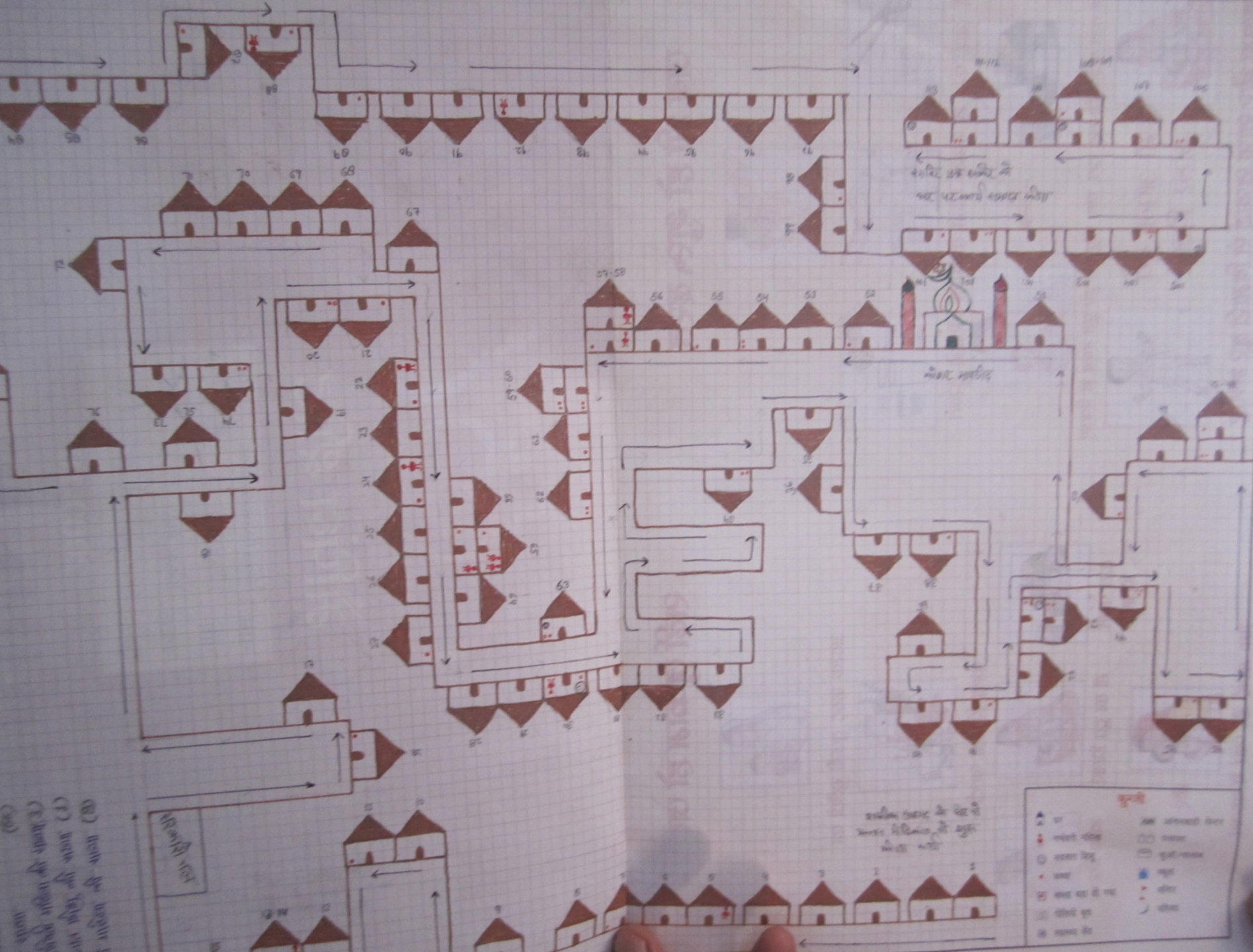
SRC – State Regional Coordinators

DMC – District Management Coordinators

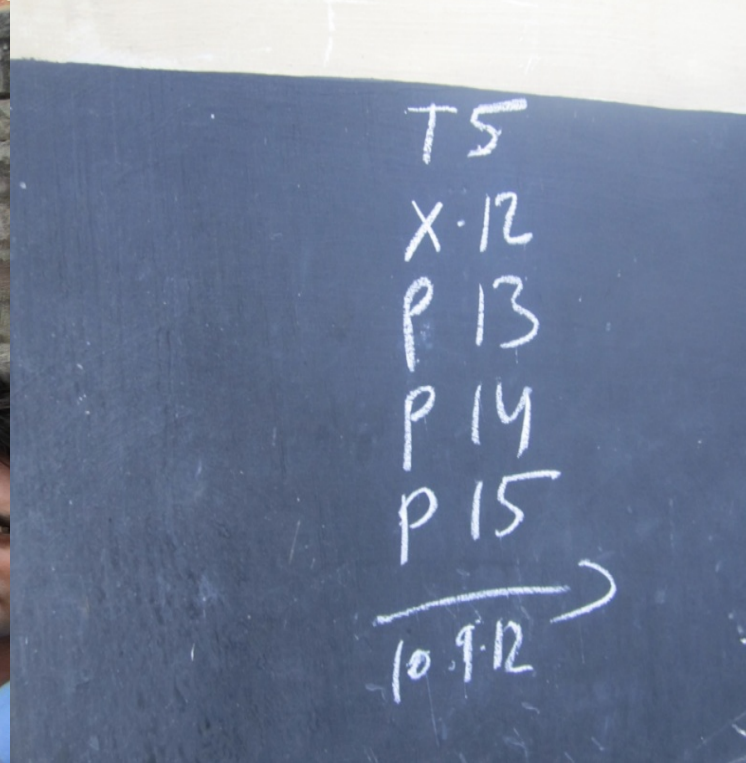
BMC – Block Management Coordinators

CMC – Community Mobilizers





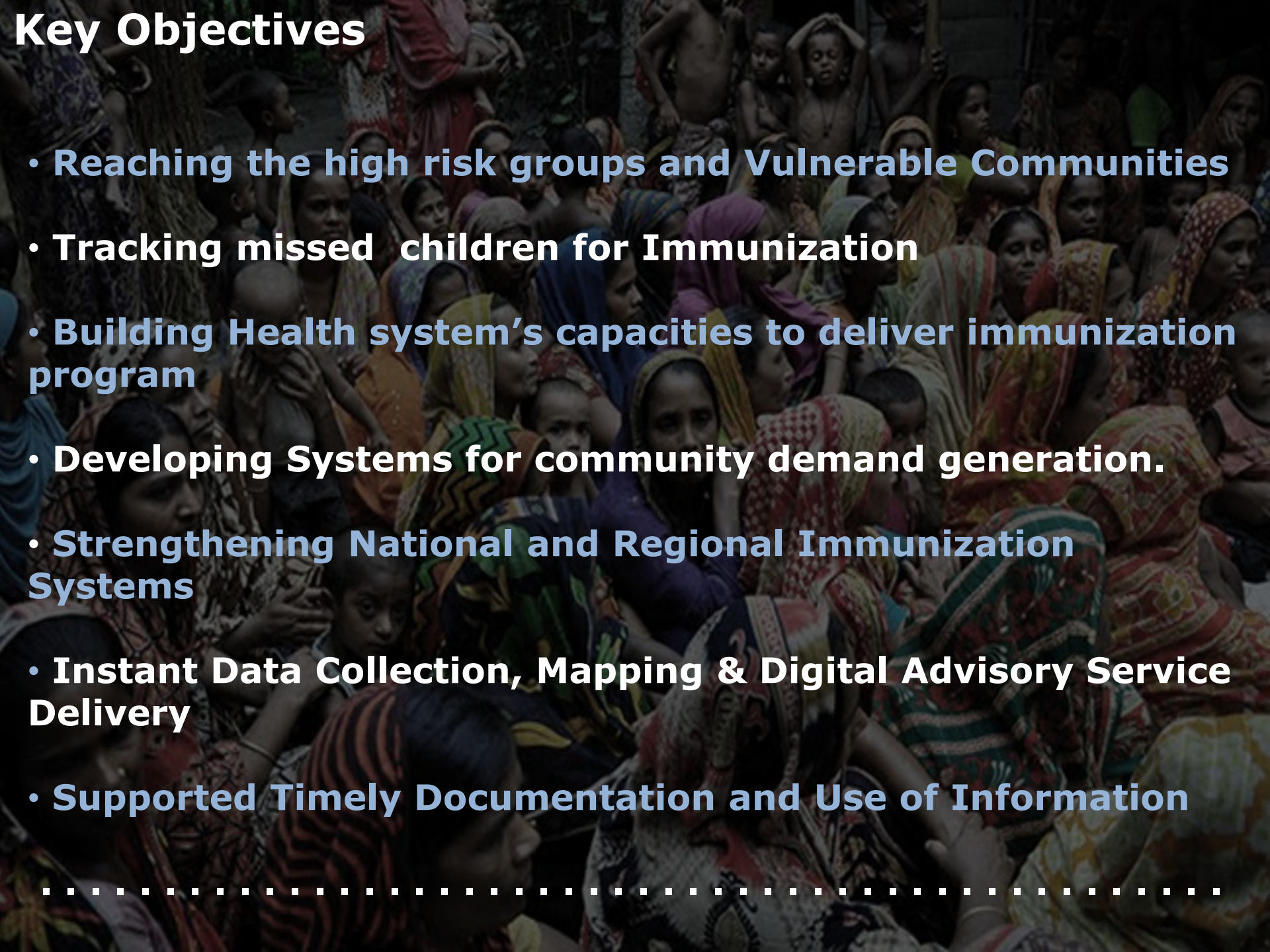




उन्हें घर की मुख्य महिला (सिलेबल)	सी.एम. सी. घर संख्या	घर के मुखिया का नाम और गर्भवती महिला का नाम	0-5 साल तक के बच्चों के नाम	लिंग	जन्म तिथि (त./माह/साल)	एस. आई. ए. र					
						11-11-10	23-1-11	21-2-11	23-3-11	24-4-11	22-5-11
3	A12	हीरे सिंह				P					
5	B12	शुविता / नरेश शांज कुमार	सनी कुमार	M	30-1-11						
2	13	सुरेश कुमार रन्जितेश	माधव ललित	F M	18-10-10 26-3-12	P	P	P X	X		
8	14	कमार उददीन श्री. सैविता	नरेश उददीन मेखल निशा	M F	21-10-10 17-11-10	P P	P P	P P	P P		X



M-Health
Solution



Key Objectives

- Reaching the high risk groups and Vulnerable Communities
- Tracking missed children for Immunization
- Building Health system's capacities to deliver immunization program
- Developing Systems for community demand generation.
- Strengthening National and Regional Immunization Systems
- Instant Data Collection, Mapping & Digital Advisory Service Delivery
- Supported Timely Documentation and Use of Information

.....

A large crowd of women and children, likely in a rural or developing area, are gathered together. Many women are wearing colorful, patterned headscarves and saris. Several children are visible, some being held by the women. The scene is dimly lit, with a dark, textured background. The word "Methodologies" is overlaid in a large, white, sans-serif font across the center of the image.

Methodologies

A large crowd of people, mostly women and children, in a rural setting. The image is dark and serves as a background for the text. The text is white and semi-transparent, allowing the background image to be visible through it. The crowd is dense, with many people looking towards the camera. The women are wearing colorful headscarves and traditional clothing. The children are interspersed among the adults, some looking directly at the camera.

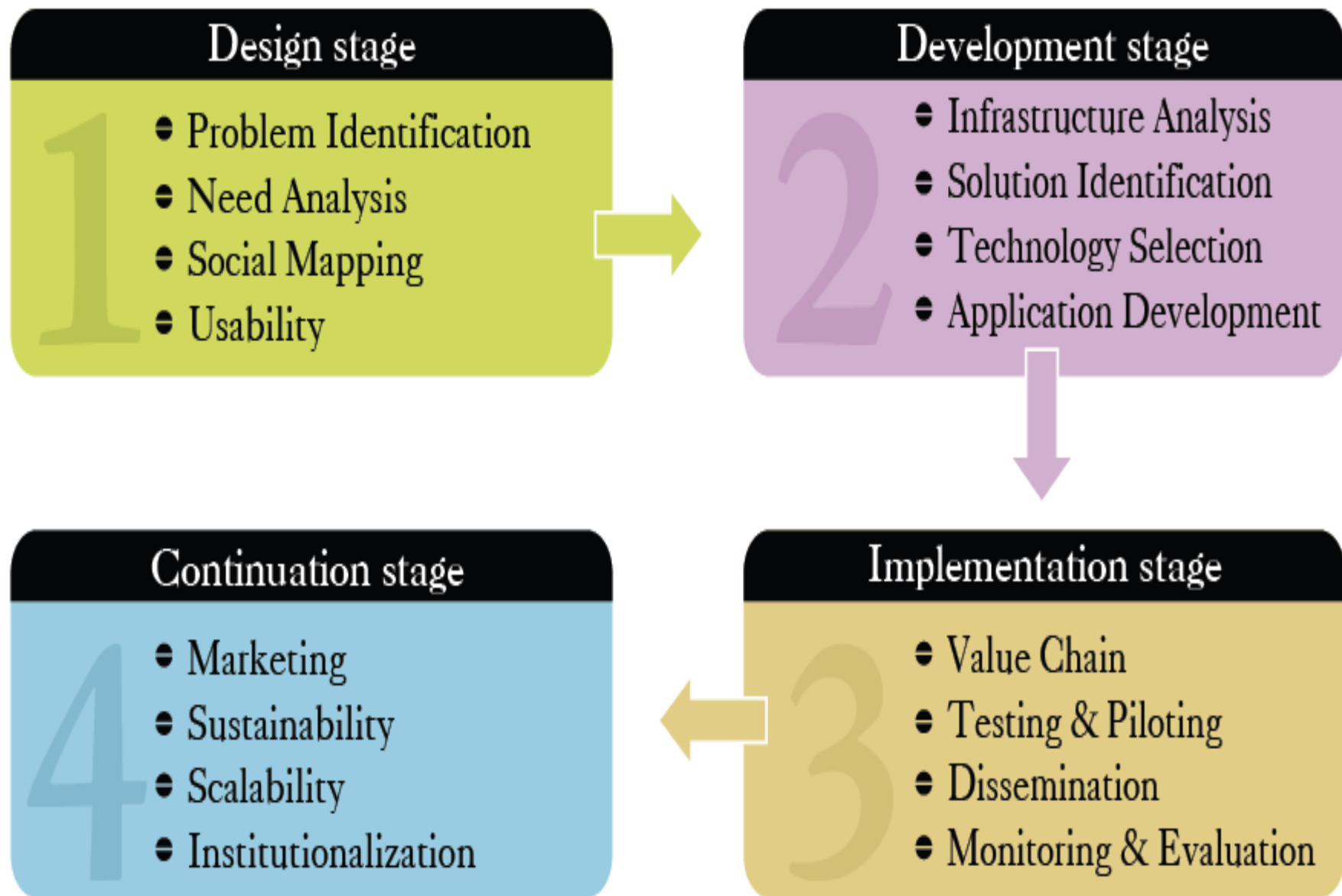
A Holistic Approach

M4D provides mobile technology solutions for social and development programs with a holistic approach.

It works towards creating a sustainable model by partnering closely with organizations, and understanding the key challenges they face in implementing the program.

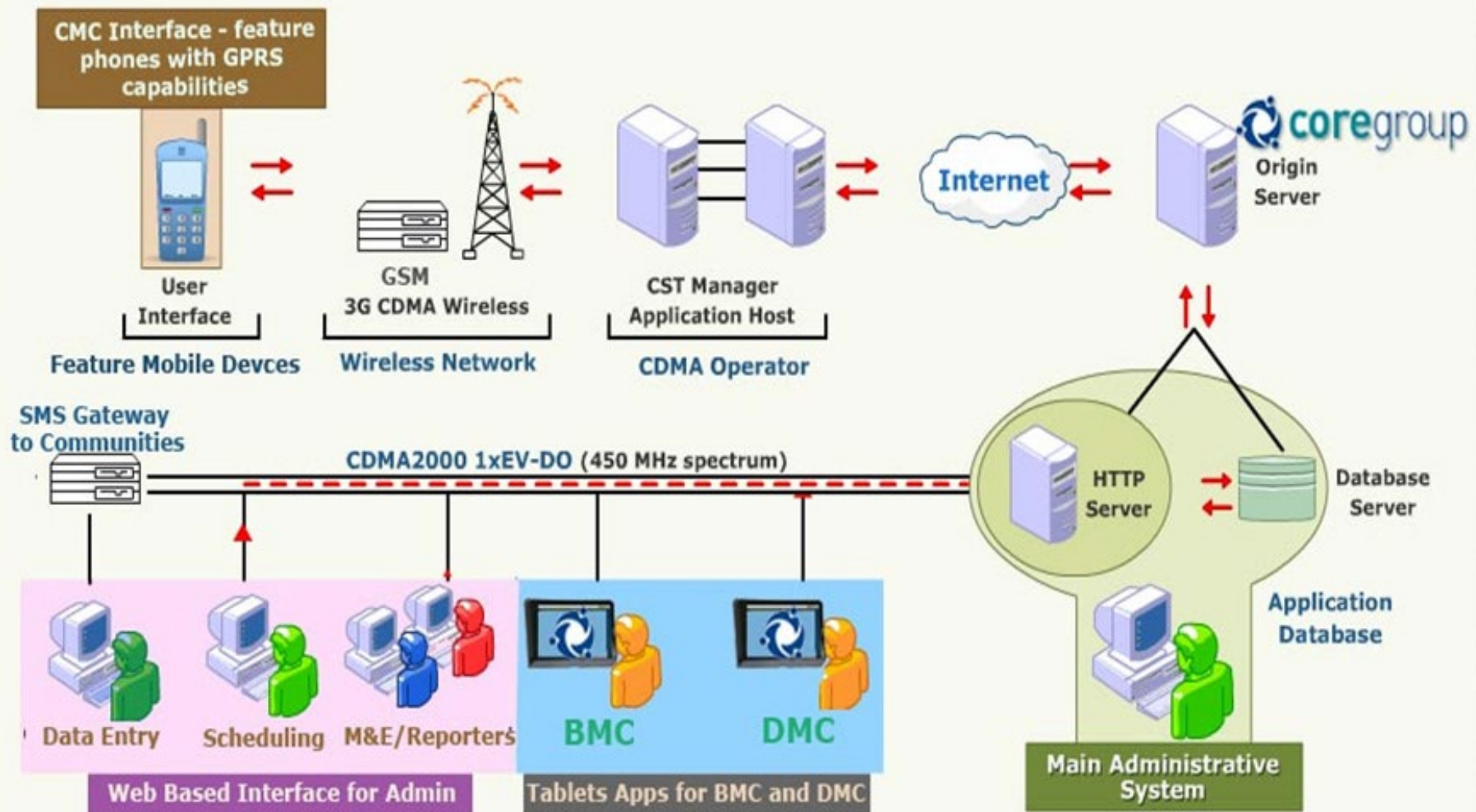
There are four stages — design, development, implementation and continuation.

Four Stages



Architecture

COREGROUP - MIS and M-Health Architecture

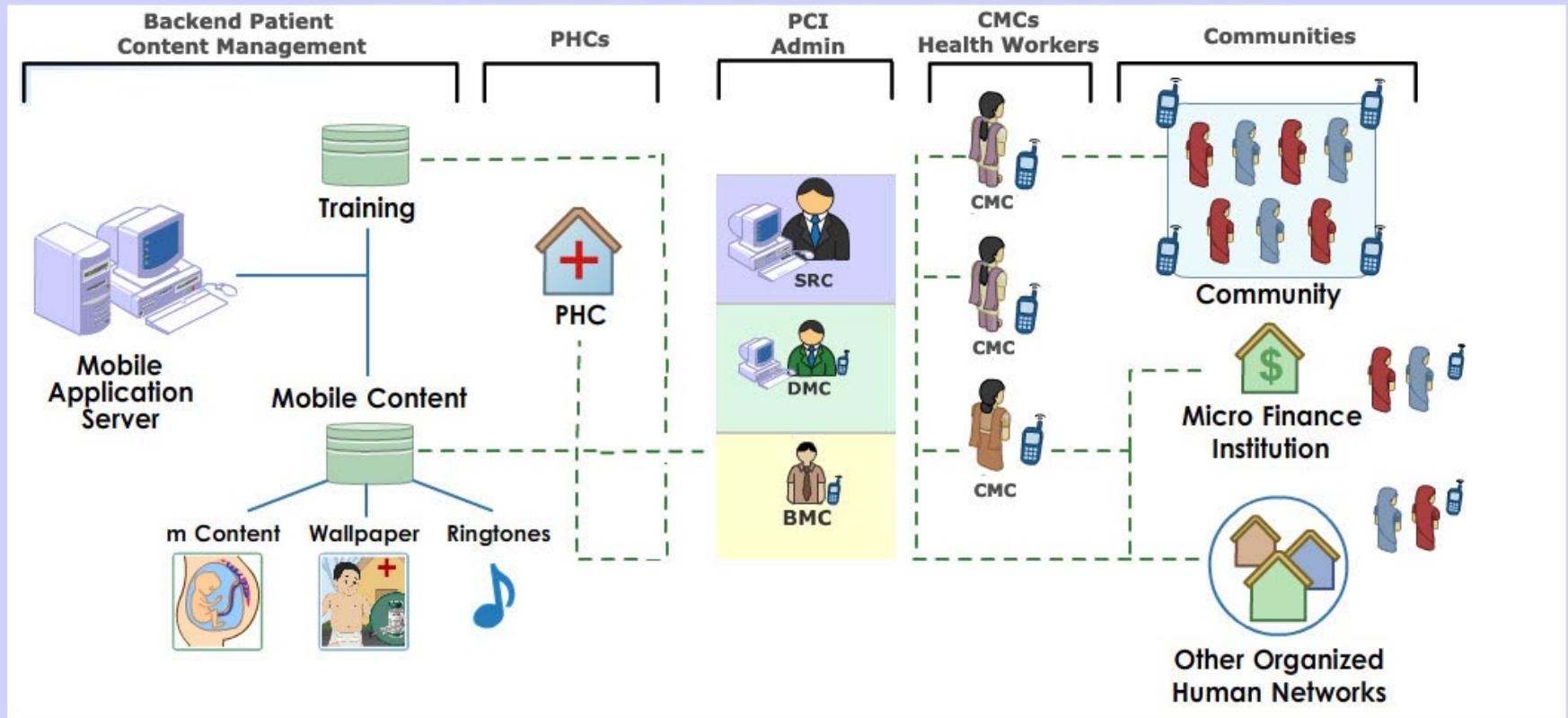


Components

- **Universal Backend System**
- **Web Application System and Platform**
- **Mobile (Feature Phone) based App. for CMCs**
- **Mobile App. (Tablet/Smartphone) for BMCs & DMCs**
- **Reporting System for Admin and Partners**
- **Training for all levels**
- **Mass Data Entry Module (Initial Phase)**
- **Mobile SMS and Application for Communities**

Universal Backend System

M-Health Services and Communication Dissemination Mechanism



Universal Backend MIS System

Establishment of Universal Hierarchical Model – State- District-Block-Cluster

Staff/Officers Registration Data Structures in Hierarchical Mode

Location Master with House-Hold Listing and Family

Mapping of Children (<5 for Polio) and others, Pregnant Women and Children who lie in range of R.I.

Universal Immunization Designer –Country and Region-wise

S.I.A System Design with Immunization Planner, Scheduler & Tracker

R.I. System Design with Immunization Planner and Tracker

All reporting Indicators with provision to add new Indicators

Report Management System

Other Relevant Data Structures



Technical Detail

Technology

Universal Backend System and Immunization Planner

Backend: Proposed System will be developed in MS-SQL technologies

Web based System: ASP.NET technologies with Java Script, AJAX Controls and XML technologies.

CMCs Mobile System: Mobile Application/Client is developed Java/J2ME

BMCs and DMCs Tablet Application: Android Platform.

A large crowd of women and children, likely in a rural or developing area, are gathered together. Many women are wearing colorful, patterned headscarves and saris. Several children are visible, some being held by the women. The scene is dimly lit, with a dark, textured background. The text "Design Views" is overlaid in the center in a large, white, sans-serif font.

Design Views

Web Application System



MANAGEMENT INFORMATION SYSTEM



Single Window Platform for all Levels of Management

User Id

Password

Sign-In

Clear

Web Application System

Super Admin Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Administrator

Company Name:- CoreGroup:Super Admin

[LogOut](#)

Company Registration	▸	Vaccine	State	District	Vaccine	▸	Company Admin	Country
Company Registration					Polio Code			
District Assign					S.I.A.Scheduler			
State Assign					SubVaccine Registration			
					Vaccine Mapping			
					Vaccine Registration			

Web Application System

Organizational Admin Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Ataur Rab

Company Name:- PCI India:SRC

[LogOut](#)

IStaff IAssign Designation IDMC Mapping IBMC Mapping IDMC Report IBMC Report

Employee General Information

Employee Name

Father Name

Mother Name

Qualification

----Select----

Phone No

Address

Sex

----Select----

Married Status

----Select----

Email Id

Joining Date

Add

Cancel

Web Application System

DMC Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Yatendra Singh Designation: DMC Organization: PCI [LogOut](#)

[IStaff](#) [IAssign Designation](#) [IBlock Creation](#) [IBlock Mapping](#) [IBMC Mapping](#) [IBMC Report](#)

Block Mapping

Country Name --Select-- State Name --Select--

District Name Block Name --Select--

Mapping

Cancel

Country Name	State Name	District Name	Block Name	Active	Edit	Delete
India	Uttar Pradesh	Meerut	ROHTA	☐	Edit	Delete
India	Uttar Pradesh	Sambhal	Naroli	☐	Edit	Delete

Web Application System

BMC Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Paveen Kusik

Designation: BMC

Organization: PCI

[LogOut](#)

[IStaff](#) [IAssign Designation](#) [ICluster Creation](#) [ICMC Mapping](#) [ICMC Report](#)

CMC-Cluster-Mapping

Block Name

Cluster/Village Name/Ward No.

CMC Name

[Assign CMC](#)

[Cancel](#)

Block Name	Cluster Name	CMC Name	isActive	Edit	Delete
ROHTA	Nagla Tashi I	Urmila	True	Edit	Delete
ROHTA	Nagla Tashi II	Amresh	True	Edit	Delete
ROHTA	Nagla Tashi III	Poonam	True	Edit	Delete

Web Application System

CMC Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Yogita

Designation: CMC

Organization: PCI

[LogOut](#)

[New Family Registration](#) [Child In Existing Family](#) [Child Details View](#) [Family Details View](#) [S.I.A.Report](#) [R.I.Report](#) [Pregnancy](#) [CMCPlan](#)

New Family Registration

CMC Area

House Details:

CMC House No.

Address/Landmark

Remarks

Family Details:

Head of Family

No of Family Members

Religion

Registration Date

Remarks

Child Details:

Number of Children(Under 5 Year)

S.No	Child Name	Gender	DOB	Father Name	Mother Name
1		---Select--			
2		---Select--			

Web Application System

CMC Operations



MANAGEMENT INFORMATION SYSTEM



Welcome Yogita Designation: CMC Organization: PCI [LogOut](#)

New Family Registration Child In Existing Family Child Details View Family Details View S.I.A.Report R.I.Report Pregnancy CMCPlan ▶

Pregnancy Registration View Pregnancy Result

CMC H.No

Pragnent Women Name

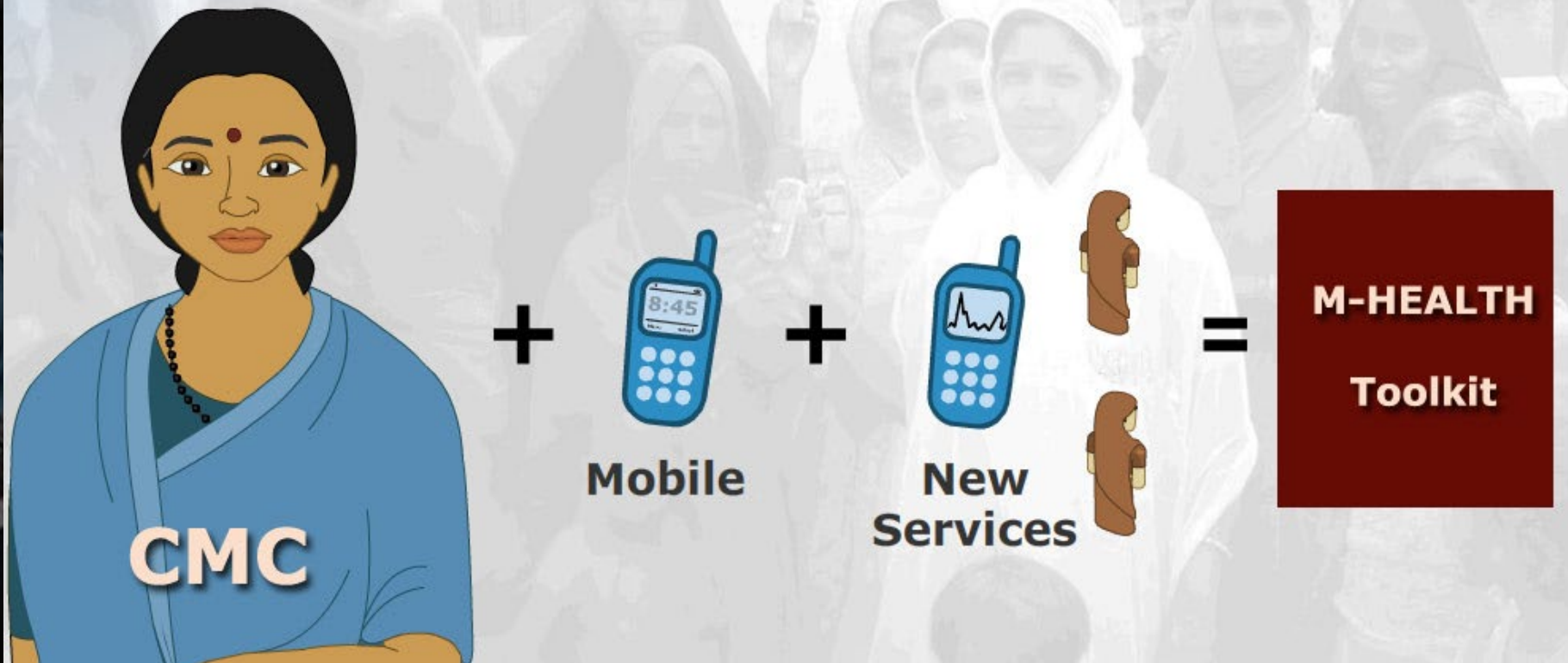
Husband Name

Expected Month of Delivery Date

Add Women

Cancel

CMCs - New Role as M-Health Worker



Mobile based App. for CMCs



Mobile based App. for CMCs



CMC – General Information Management

New House, Family and Child Registration

Pregnant Woman Registration and Status Tracking

Community Social Mobilization Register

Training and Educational toolkit for CMCs

Remote CMC Channel

CMC Knowledge Channel

Mobile based App. for CMCs

S.I.A. Operations



The screenshot shows a mobile application interface with a yellow header bar labeled "Menu". Below the header is a central black box with white text listing five options: "General Information", "New Registration", "RI Status", "Pregnant Women Status", and "SIA Round". The "SIA Round" option is highlighted with a yellow rectangular border. The background of the app interface is a brown grid pattern. On either side of the central black box are two vertical yellow oval frames containing small images of people. At the bottom of the app interface are two yellow buttons labeled "Select" and "Exit".

Menu

General Information

New Registration

RI Status

Pregnant Women Status

SIA Round

Select Exit

Mobile based App. for CMCs

S.I.A. Operations

SIA Round

Request Data

Child Track Sheet

X Childs

Submit Today Data

Sumitted X Childs

Select

Back

Mobile based App. for CMCs

S.I.A. Operations

Request Data

Polio Day

3

Starting Round H No

26

Ending Round H No

230

(≤ 387)

Team Number

1232

Enter

Submit

Back

Mobile based App. for CMCs

S.I.A. Operations

SIA Round

Request Data

Child Track Sheet

X Childs

Submit Today Data

Sumitted X Childs

Select

Back

Mobile based App. for CMCs

S.I.A. Operations

Child Track Sheet

CMC House No	23
Child Name	Guddu
Family Head	Mursaleem
Fathar Name	Mursaleem
Status	P

Enter

Save

Back

Mobile based App. for CMCs

S.I.A. Operations

SIA Round

Request Data

Child Track Sheet

X Childs

Submit Today Data

Sumitted X Childs

Select **Back**

Mobile based App. for CMCs

S.I.A. Operations

Child Track Sheet

CMC House No	17
Child Name	Kaif
Family Head	Deen Mohd.
Fathar Name	Deen Mohd.
Status	XS

Enter

Save

Back

Mobile based App. for CMCs

S.I.A. Operations

SIA Round

Request Data

Child Track Sheet

X Childs

Submit Today Data

Sumitted X Childs

Select

Back

Mobile based App. for CMCs

S.I.A. Operations

Submit Data			
P	6	Pb	0
XL	4	XS	1
XV	1	XH	1
XR	0	XL-P	0
XL-Pb	0	XS-P	0
XS-Pb	0	XV-P	0
XV-Pb	0	XH-P	0
XH-Pb	0	XR-P	0
XR-Pb	0	All Childs	13
Send		Back	

A large crowd of women and children, many wearing colorful headscarves, gathered outdoors. The image is dark and serves as a background for the text.

Mobile based App. for CMCs

S.I.A. Operations

Phase II Activity with X houses

Mobile based App. for CMCs

S.I.A. Operations

List of X House		
CMC H No	Child	Status
25	---	XV
24	Rabia	XL
23	Rahim	XH
21	Guddu	XL
21	Simran	XL
19	---	XL
17	Gudiya	XS
Select		Back

Mobile based App. for CMCs

S.I.A. Operations

Change Status	
CMC House No	25
Child Name	---
Family Head	Shakeel
Fathar Name	---
Ist Visit Status	XV
Biphase Status	P
Enter	
Back	

Mobile based App. for CMCs

R. I. Operations

Menu

General Information

New Registration

RI Status

Pregnant Women Status

SIA Round

Select

Exit

Mobile based App. for CMCs

R. I. Operations

RI Status

[View Child Vaccination Detail](#)

List of the due Vaccination Child

Select

Back

View Child Vaccination Deatail

Searching Option

House No

Enter Text

53

Child Id

79

Child Name

Shanu

Due Vaccination Name OPV1

Due Vaccination Date 01/22/2020

[All Vaccination List](#)

Back

Mobile based App. for CMCs

R. I. Operations

RI Status

View Child Vaccination Detail

List of the due Vaccination Child

Select**Back**

Child Vaccination List

Zero doz	☐	OPV3	☐
BCG	☐	DPT3	☐
OPV1	☐	Measles	☐
DPT1	☐	Vitamin A	☐
OPV 2	☐	Booster	☐
DPT 2	☐		

☐ Pending ☐ Recived ☐ Due ☐ N.A.

Due Date: 02/07/2013

Back

Mobile based App. for CMCs



CMC – General Information Management

New House, Family and Child Registration

Pregnant Woman Registration and Status Tracking

Community Social Mobilization Register

Training and Educational toolkit for CMCs

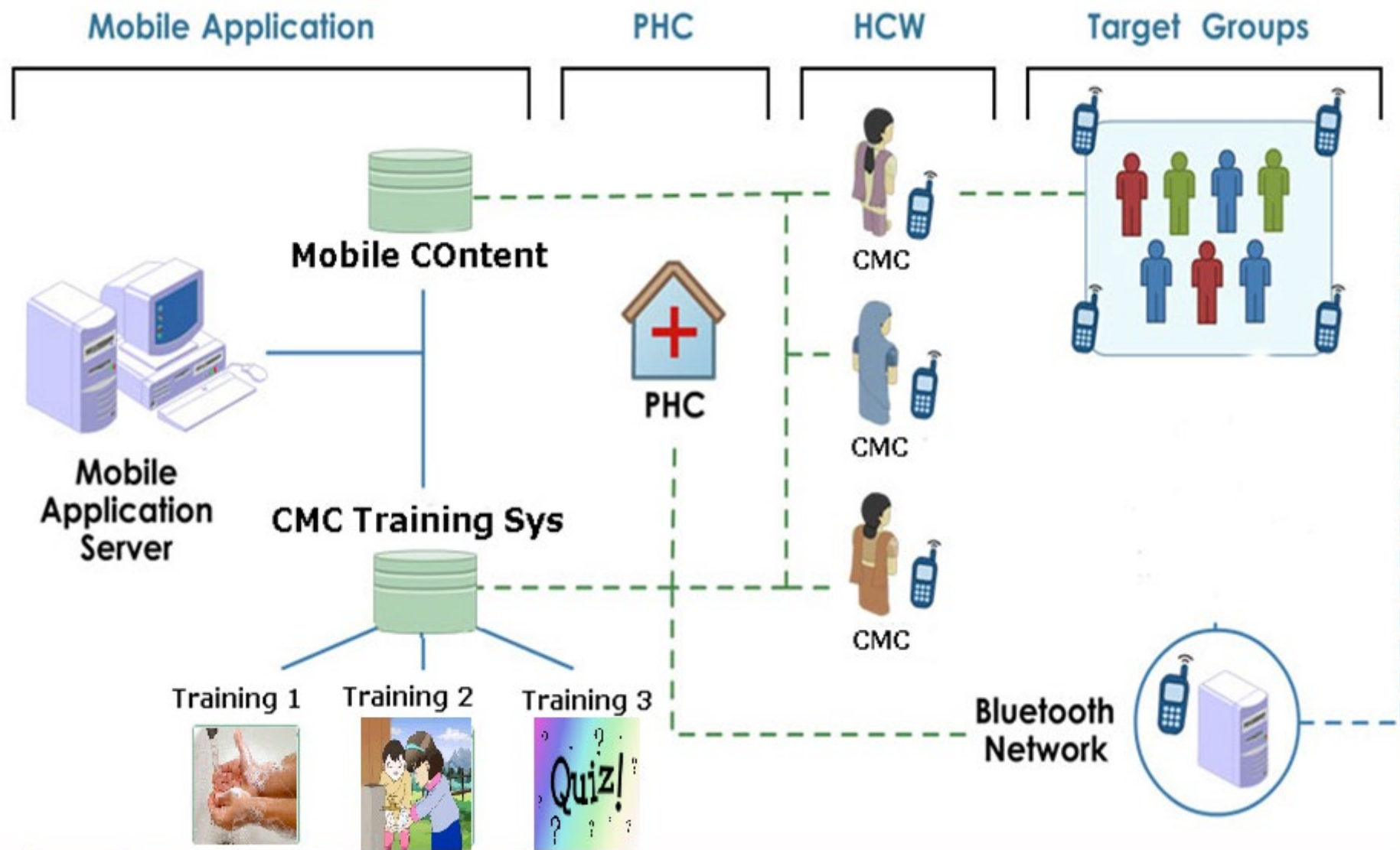
CMC Knowledge Channel

Training of Staff & Community



Training of
CMCs
Community Members

CMC Trainer: Architecture



Mobile based App. for CMCs



CMC – General Information Management

New House, Family and Child Registration

Pregnant Woman Registration and Status Tracking

Community Social Mobilization Register

Training and Educational toolkit for CMCs

CMC Knowledge Channel

Mobile based App. for CMCs

CMC Knowledge Channel

How to wash hands properly

Animations

Quiz

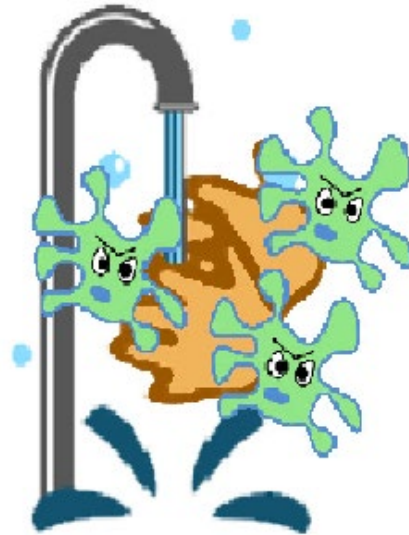
e-help and Learn

Select

Exit

How to Wash Your Hands

Step 1/6



1. Wet hands

Select

Exit

Mobile based App. for CMCs

How to Wash Your Hands

Step 2/6



2. Soap

Select

Exit

How to Wash Your Hands

Step 3/6



3. Wash at least
20 seconds

Select

Exit

How to Wash Your Hands

Step 4/6



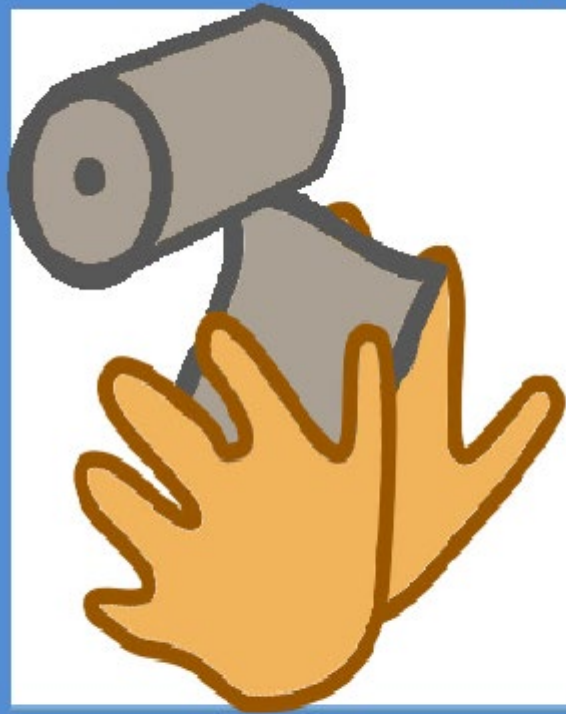
4. Rinse

Select

Exit

How to Wash Your Hands

Step 5/6



5. Dry Hands

Select

Exit

How to Wash Your Hands

Step 6/6



**6. Turn off water
with a paper towel**

Select

Exit

Mobile based App. for CMCs

CMC Knowledge Channel

How to wash hands properly

Animations

Quiz

e-help and Learn

Select

Exit

How to Wash Your Hands



Select

Exit

Mobile based App. for CMCs

CMC Knowledge Channel

How to wash hands properly

Animations

Quiz

e-help and Learn

Select

Exit

How to Wash Your Hands

Question 1/5

How long should you wash your hands?

- ☐ 5 to 10 seconds
- ☐ 15 to 20 seconds
- ☐ 60 - 90 seconds

NEXT ►

Select

Exit

Mobile based App. for CMCs

How to Wash Your Hands

Question 2/5

T/F: You should wash your hands after petting a dog or cat.

- ☐ True
- ☐ False

NEXT ►

Select

Exit

How to Wash Your Hands

Question 3/5

T/F: Fewer people are currently washing their hands after using the restroom than in 2006.

- ☐ True
- ☐ False

NEXT ►

Select

Exit

How to Wash Your Hands

Question 4/5

According to the Centers for Disease Control, which is the single most important way to prevent the spread of illness, including colds and flu?

- ☐ Covering your mouth when you cough.
- ☐ Sneezing into a tissue.
- ☐ Washing your hands regularly throughout the day.
- ☐ Getting a flu vaccine.

NEXT ►

Select

Exit

How to Wash Your Hands

Question 5/5

T/F: Antibacterial soaps kill more germs than regular soaps do.

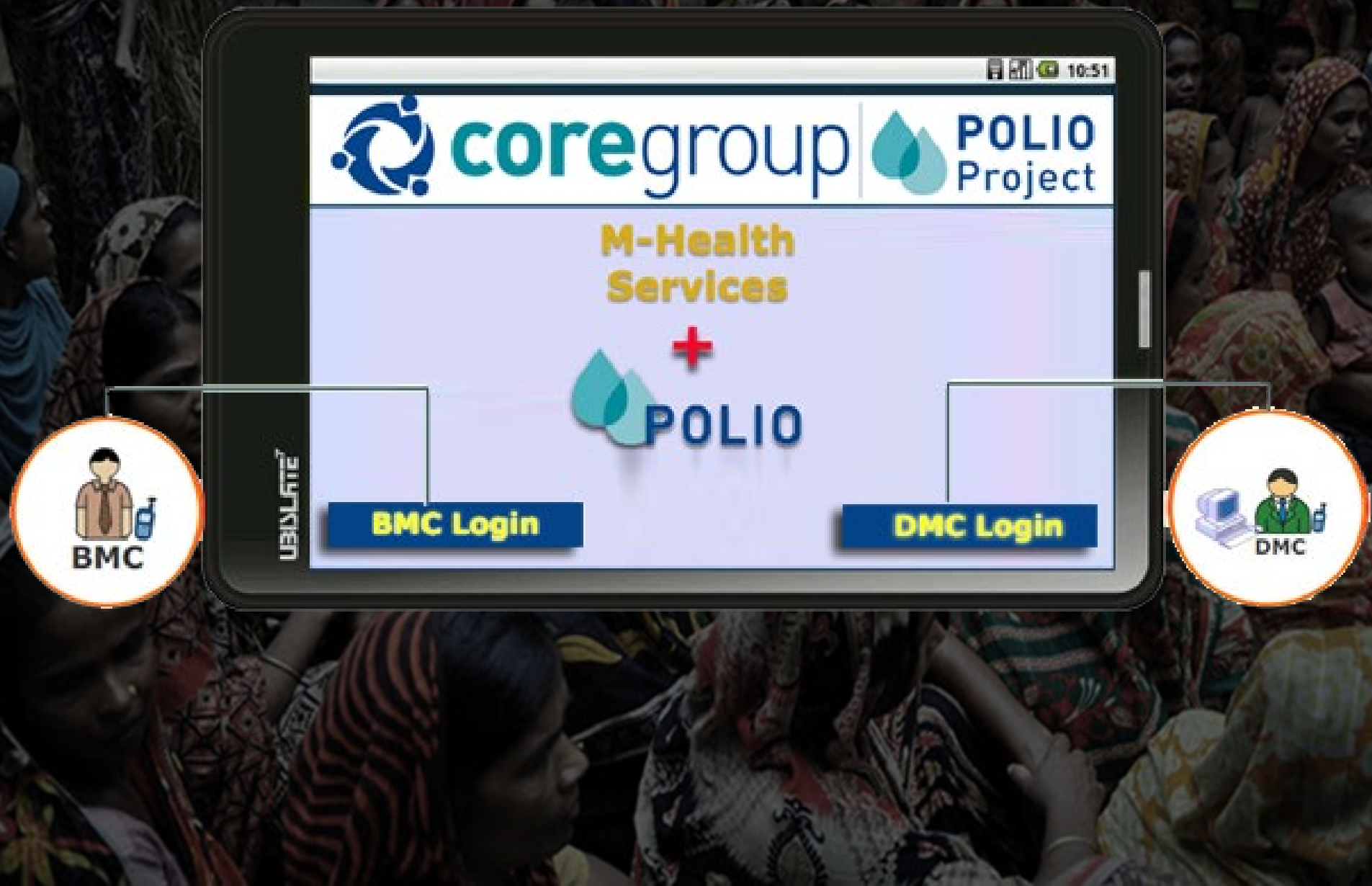
- ☐ True
- ☐ False

NEXT ►

Select

Exit

Tablet App. for BMCs & DMCs



Tablet App. for BMCs & DMCs

DMC Operations



Tablet App. for BMCs & DMCs

DMC Operations

3G 10:53 AM	
PolioProject	
FORMAT-8	
February	
CONTENT	VALUES
CMC Name	CMC 3
Ward	Ward 3
Block	Block 5
Total Child	400
Total No of houses	220
0-5 year child	375
From October 2011 to Till Now	567
From October 2010 to September 2011	690
From October 2009 to September 2010	730
From October 2008 to September 2009	410
From October 2007 to	320

Tablet App. for BMCs & DMCs

BMC Operations

3G 10:42 AM

PolioProject

Username

Password

Login Clear All Back

3G 10:44 AM

PolioProject

CMC List

Add New CMC

Assign Area to CMC

Create New Area

CMC Report

Back

Tablet App. for BMCs & DMCs

BMC Operations

3G 10:44 AM

PolioProject

EMPLOYEE REGISTRATION FORM

Employee Name

Employee's Father Name

Employee Mother's Name

Select Education

Employee's Phone Number

3G 10:45 AM

PolioProject

CMC Area Registration
Cluster Creation

Company Name :

Village/Ward No/Area

Total Populatipn

Hindu Population

Muslim Population

☒ HRA

Tablet App. for BMCs & DMCs

BMC Operations

3G 10:45 AM

PolioProject

CMC REPORT

JANUARY 2013

NAME	TNOH	P0	X	X-P
CMC 1	500	30	10	20
CMC 2	451	12	23	24
CMC 3	350	34	21	17
CMC 4	120	33	31	3
CMC 5	400	22	21	7
CMC 6	502	12	21	9
CMC 7	532	10	23	12
CMC 8	412	22	11	16
CMC 9	552	21	9	9
CMC 10	522	23	10	12
CMC 11	459	21	7	11
CMC 12	470	12	3	5

3G 10:54 AM

PolioProject

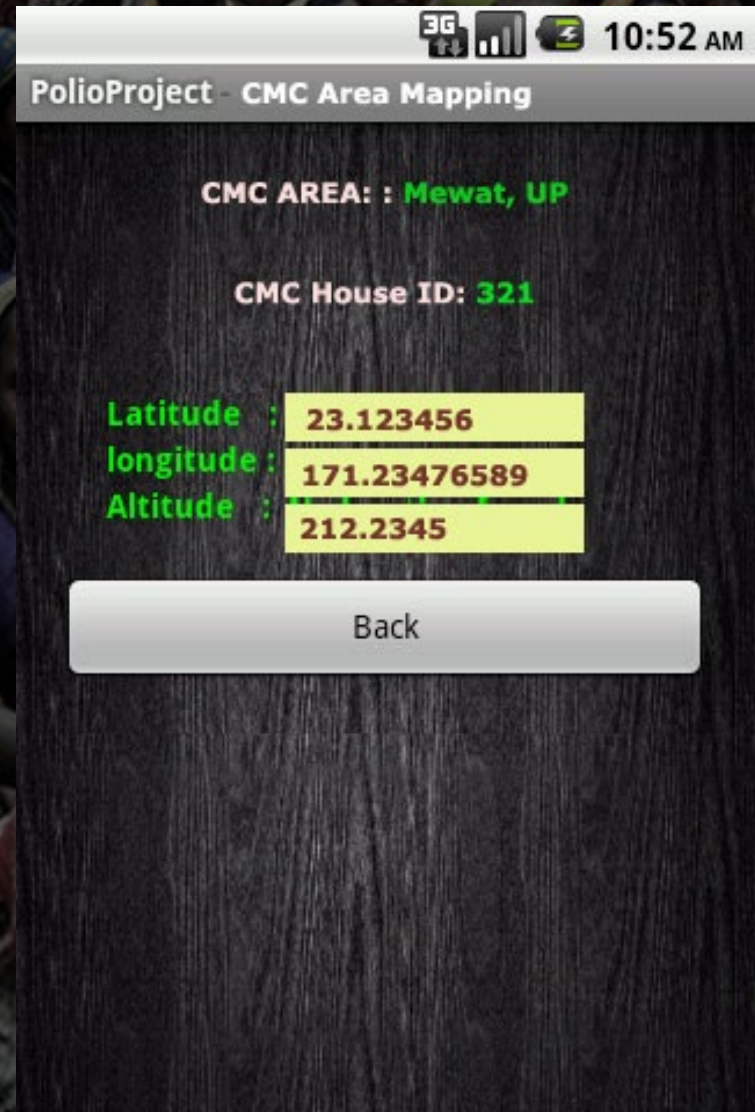
FORMAT-8

February

CONTENT	VALUES
Total dead Child	2
Total survived Child	38
Round 0 Doze	40
0 Doze	32
BCG	32
DPT-1	31
OPV-1	24
DPT-2	20
OPV-2	25
DPT-3	24
OPV-3	23
Khasra	32
Vitamin A	31
Booster	42

Tablet App. for BMCs & DMCs

CMC Operations – Area Mapping



Reporting System

CHILD S.I.A. ROUND MANAGEMENT SYSTEM

Submit Data

Survey From To

Select Child Polio Immunization During Rounds
 Fill All S.I.A Dates (16 Dates in Register 2011-2012)

Family No.	Head Of Family/Pregnant Women	Child Name(0-5)	Gender	DOB	10/7/2011	11/8/2011	1/9/2012	2/6/2012	3/5/2012	4/9/2012
6	R>K.Arora	Sanjay	Male	Dec 17 201	P	P	P	P	P	P
6	R>K.Arora	Sanjana	Female	Dec 12 201	X	X-P	X	P	P	P
7	Arshad Khan	Sumbul	Female	Nov 26 200	X	X	X	X	X	X
7	Arshad Khan	Nadeem	Male	Nov 27 201	X	X	X	X	X	X
7	Arshad Khan	Sumbul	Female	Nov 26 200	P	P	P	P	P	P
7	Arshad Khan	Nadeem	Male	Nov 27 201	P	P	P	P	P	P
7	Arshad Khan	Sana	Female	Feb 8 201	P	P	P	P	X	P
8	Z M Quraishi	Zaheer	Male	Jan 5 200	P	P	P	P	P	P
8	Z M Quraishi	Faisal	Male	Feb 24 200	P	P	P	P	P	P
8	Z M Quraishi	Masood	Male	May 6 201	P	P	P	P	P	P
8	Z M Quraishi	Nuha	Female	Apr 11 201	P	X	X	P	X	P

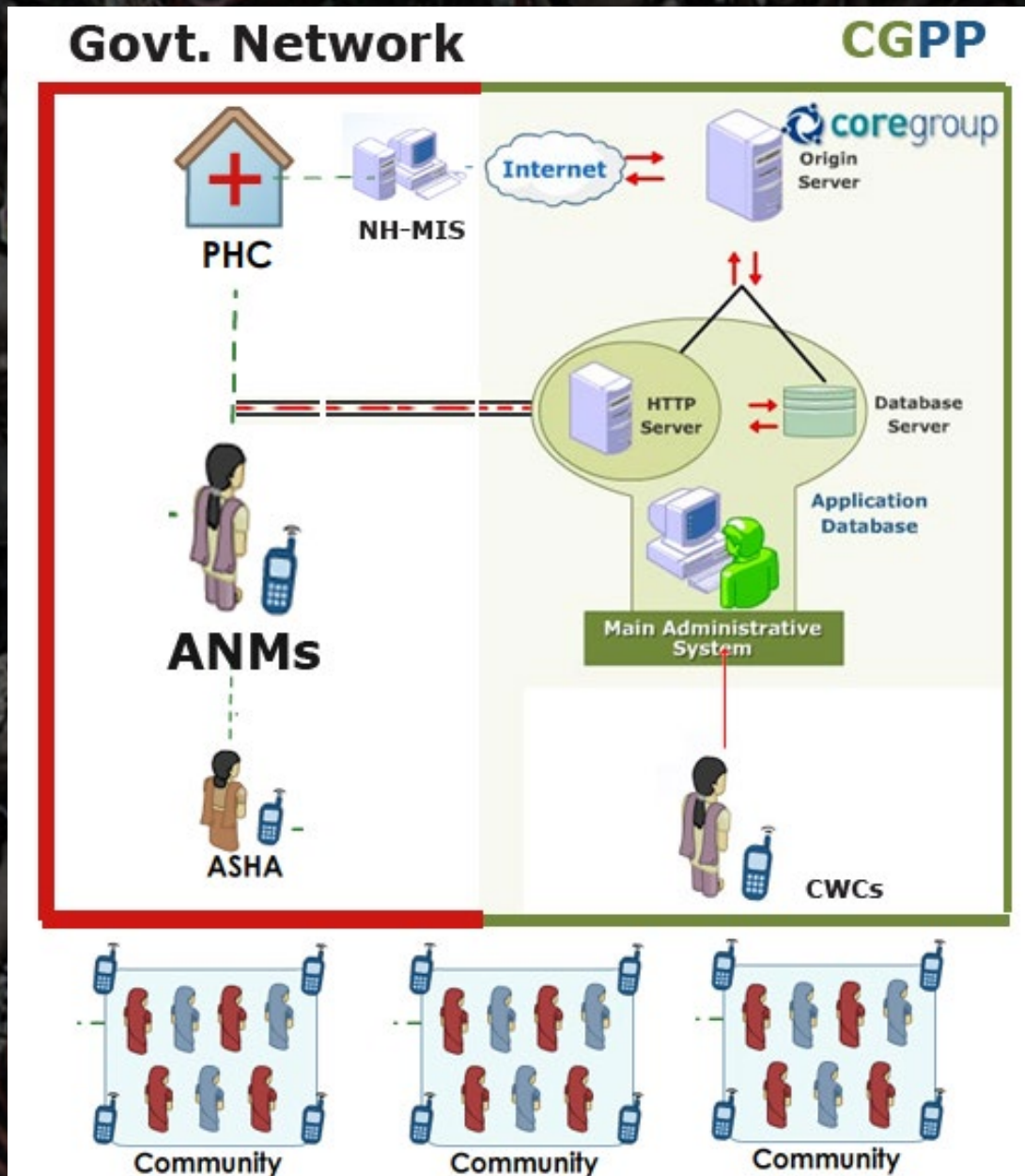


The Future

The Future And Scope Of My Project

- ❑ Integrator with National Health
- ❑ GPS Mapping of Villages for Surveillance for High Risk, other Behavioral Patterns and GIS Data Analysis
- ❑ Mobile Training Kit for CMCs
- ❑ Mobile App Center for Community Programs like Mothers Meetings , general awareness programs.
- ❑ Scaling the Project for Universal Immunization **Globally**
- ❑ Developing Community level Digital Tools – for SMS, Feature phone and Smartphone based
- ❑ Gaming with Live Data for Training and Simulations

Integrator with National Health



GPS/GIS Data Analysis

CMC Name: Kamala

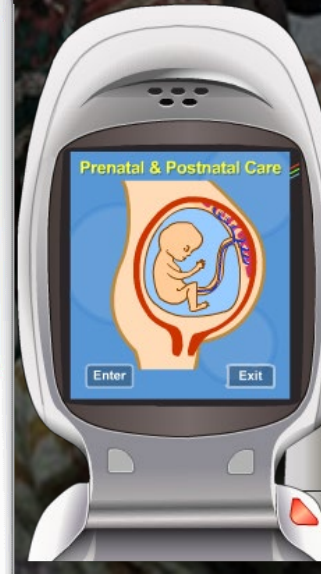
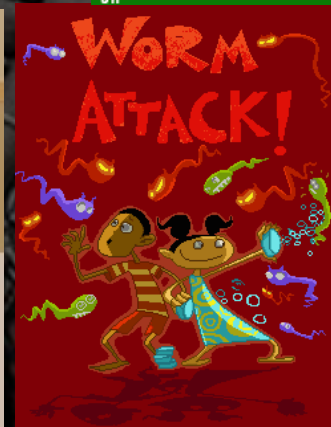
RISK PATTERNS

CMC AREA: Nagla Tashi II, Meerut, UP

Pop: 14231
Children<5: 258
Vacc: 53



Community Toolkit



टी.बी. के रोगाणु धीरे-धीरे मरते हैं।
इनको मारने के लिए 6-8 महीने
तक नियमित रूप से दवा लेनी
चाहिए।

Simulation/Games with Live Data



A large crowd of women and children, likely in a rural or developing area, are gathered together. Many women are wearing colorful, patterned headscarves and saris. Several children are visible, some being held by the women. The scene is dimly lit, with a dark, textured background. The word "CODING" is superimposed in the center in a large, white, serif font.

CODING

Coding

MyCanvas.java

```
import java.io.IOException;
import javax.microedition.lcdui.Canvas;
import javax.microedition.lcdui.Graphics;
import javax.microedition.lcdui.Image;
import javax.microedition.midlet.MIDlet;
public class MyCanvas extends Canvas{
    int W=getWidth(),H=getHeight(),index=0;
    Image myimage;
    private MIDlet midlet;
    Image image[]=new Image[8];
    MyCanvas()
    { try {
        image[0] = Image.createImage("/1.png");
        image[1] = Image.createImage("/2.png");
        image[2] = Image.createImage("/3.png");
        image[3] = Image.createImage("/4.png");
        image[4] = Image.createImage("/5.png");
        image[5] = Image.createImage("/6.png");
```


CONT...

```
myimage=image[0];  
    } catch (IOException ex) {  
        ex.printStackTrace();  
    }  
protected void paint(Graphics g) {  
    // g.setColor(255,0,0);  
    // g.fillRect(0, 0, W, H);  
    g.setColor(0,0,0);  
    g.fillRect(0, 0, W, H);  
    g.drawImage(image[index],  
getWidth()/2,getHeight()/2,  
Graphics.VCENTER|Graphics.HCENTER);  
    g.setColor(255,0,0);  
    g.drawString("Next",W-25 ,H-20,0);  
}  
protected void keyPressed(int keycode) {  
    if(keycode==-7)  
    {  
        if(++index>7)index=0;  
        repaint();    } } }
```


Coding

Midlet.java

```
import javax.microedition.midlet.*;
import javax.microedition.lcdui.*;

public class Midlet extends MIDlet {
    public void startApp() {
        Display.getDisplay(this).setCurrent(new MyCanvas());
    }

    public void pauseApp() {
    }

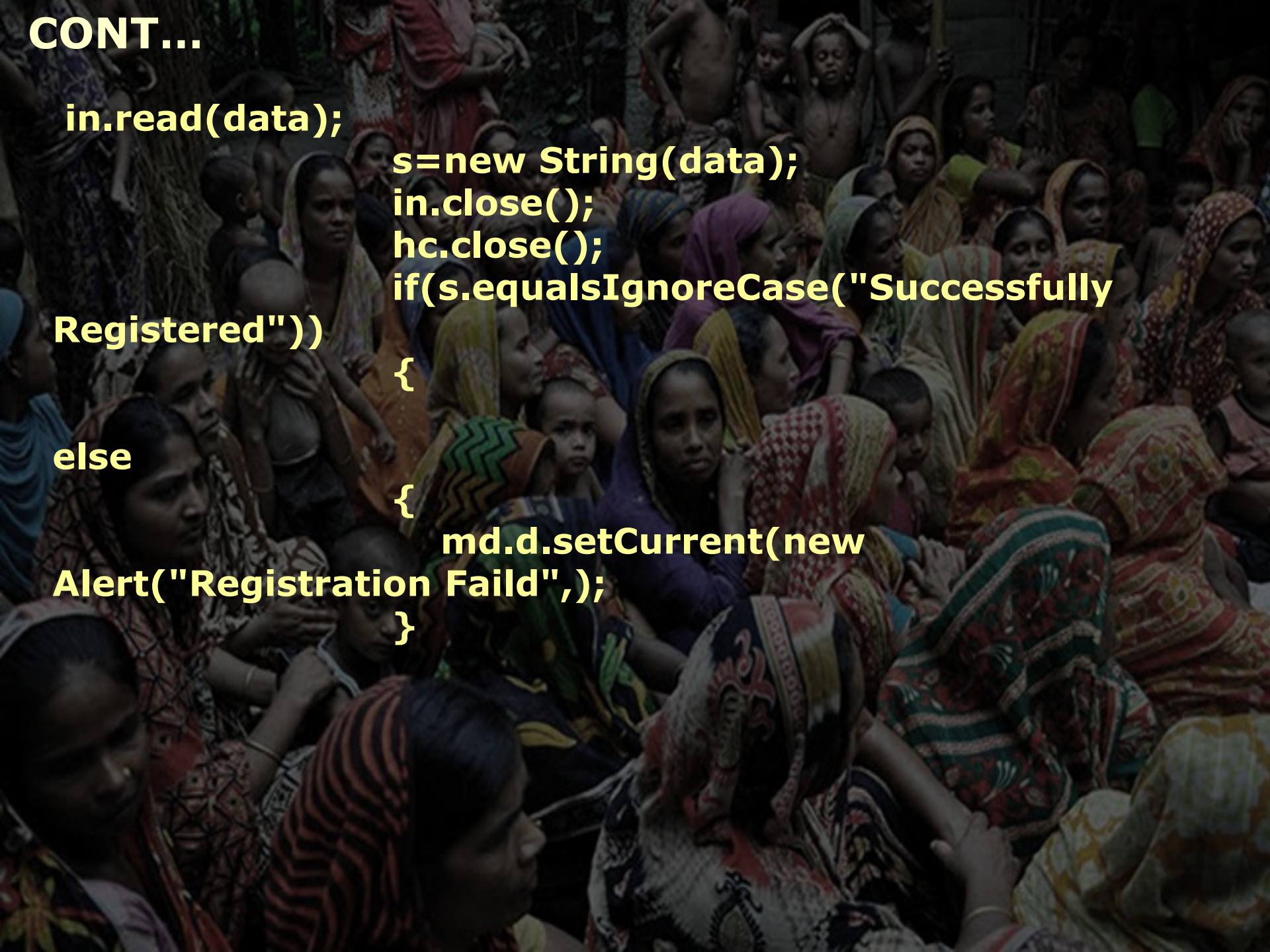
    public void destroyApp(boolean unconditional) {
    }
}
```


Calling services

```
if(c==submit)
{
    setString();
    final String url =
"http://192.168.1.201/nikhil/Registration.aspx?uid="+use
rid+" &password="+password;
    new Thread(new Runnable(){
        public void run() {
            try {
                HttpURLConnection hc =(HttpURLConnection)Connector.open(url);
                DataInputStream in;
                String s;
                byte[] data = null;
                int length = (int)hc.getLength();
                data = new byte[length];
                in = new
                DataInputStream(hc.openInputStream());
```


CONT...

```
in.read(data);  
s=new String(data);  
in.close();  
hc.close();  
if(s.equalsIgnoreCase("Successfully  
Registered"))  
{  
else  
{  
md.d.setCurrent(new  
Alert("Registration Faild",);  
}
```



A large crowd of women and children, likely in a rural or developing area, are gathered together. Many women are wearing colorful, patterned headscarves and saris. Several children are visible, some being held by the women. The scene is dimly lit, suggesting an indoor or shaded outdoor environment. The word "Thanks" is overlaid in the center in a bold, orange font.

Thanks